

Sketching When an AI Tutor Should Abstain: A Study Protocol for Concept-Based Design of Abstention in Retrieval-Augmented Question Answering

Yeoeun Choi

Independent Researcher, Boston, MA, USA

ychoi0908@gmail.com

Study protocol. No results yet. Not peer reviewed.

Abstract

In earlier work, I built a retrieval-augmented question-answering system from standard components and found that it answered all 258 unanswerable questions in a 500-question SQuAD 2.0 sample, in the same form as its correct answers. A tutor that never declines to answer gives learners no cue for when to doubt it. Abstention is usually handled as a scoring problem: a model or calibrator outputs one confidence value and the system answers when it is high enough. This protocol describes a study that treats abstention instead as a design problem. Using model sketching, a method for composing functional models from concepts people can read, I will author concepts that describe why a tutor should decline (for example, “the retrieved passage does not mention the question’s main entity”), combine them into small sketch models, and compare them with a retrieval-score threshold on the same 500 questions and on a held-out sample of 200. A dated design journal records how sketching changes the way I frame the problem. The study reuses existing data and involves no participants. I state the research questions, procedure, analysis plan, and limitations before collecting any results.

Keywords: abstention; model sketching; retrieval-augmented generation; unanswerable questions; trust calibration; AI tutors

1. Introduction

Students increasingly ask AI systems questions they cannot yet answer themselves. In that situation, a learner can only calibrate trust if the system gives some signal of when it has no basis for an answer. My earlier study of a retrieval-augmented question-answering (RAG) system found that this signal was missing entirely: the system answered every unanswerable question it was given (Choi, 2026).

The usual remedy is a better score. A calibrator estimates the probability that an answer is correct, and the system abstains below a threshold (Kamath et al., 2020). That can work, but a single number hides *why* the system should not answer, and the reason is exactly what an honest tutor could tell a learner. This study asks whether the reasons for abstaining can be designed directly, as named concepts, using model sketching (Lam et al., 2023).

Can naming the reasons an AI tutor should decline to answer show what a single confidence score hides?

This document is a protocol. It fixes the research questions, procedure, and analysis before any sketching begins, so that the eventual report can be checked against what I planned.

2. Background

2.1 Abstention in question answering

SQuAD 2.0 added over 50,000 unanswerable questions, written by crowdworkers to look similar to answerable ones, so that systems must “determine when no answer is supported by the paragraph and abstain from answering” (Rajpurkar et al., 2018). Kamath et al. (2020) studied selective question answering under domain shift and found that abstention policies based only on a model’s softmax probabilities fare poorly, because models are overconfident on out-of-domain inputs. A separately trained calibrator did better: at 80% accuracy it answered 56% of questions, compared with 48% when using the model’s probabilities directly.

2.2 Model sketching

Lam et al. (2023) observed that machine learning practitioners often tunnel on low-level technical details such as architectures and metrics. They introduced model sketching, in which practitioners compose high-level, human-understandable concepts into quick functional approximations of a model. Their Python package, ModelSketchBook, scores each concept zero-shot (for text, by prompting GPT-3) and combines concept scores with simple aggregators such as logistic regression or a decision tree. In an evaluation with 17 practitioners, sketching shifted attention from implementation to higher-level exploration and helped participants identify gaps in the problem formulation.

2.3 The gap

Table 1 summarizes how these lines of work relate to this study. I am not aware of work that uses model sketching to design when a learner-facing system should decline to answer.

Table 1. Prior work and what this study adds.

Work	What it does	What it leaves open
Rajpurkar et al. (2018)	Adds unanswerable questions to SQuAD	A benchmark, not a design method
Kamath et al. (2020)	Trains a calibrator that decides when to abstain	The decision is one score, not reasons people can inspect
Karpukhin et al. (2020)	Dense passage retrieval, used in my pipeline	Retrieving a passage is not the same as having a basis to answer
Lam et al. (2023)	Composes readable concepts into quick models	Evaluated on tasks such as content moderation, not abstention or education

3. Research questions

1. **RQ1.** Which human-understandable concepts separate the questions the system should have declined from those it should have answered?
2. **RQ2.** How do concept-based sketch models of abstention compare with a single retrieval-score threshold on the same questions?
3. **RQ3.** What does the sketching process reveal about the abstention problem that the original metrics (exact match, F1, Recall@K) did not?

4. Method

4.1 Data and system

The study reuses the evaluation run from my earlier work; no new data are collected. The question set is a fixed, seeded sample of 500 questions from the SQuAD 2.0 validation split: 242 answerable and 258 labeled unanswerable. The retriever is Dense Passage Retrieval (Karpukhin et al., 2020) with the publicly released Natural Questions checkpoints, searched through an exact inner-product FAISS index over about 29,000 chunks drawn from about 20,000 SQuAD 2.0 passages. The generator is Flan-T5-base, prompted

with no instruction about unanswerable questions. For each question, the study needs the question, the top retrieved passages and their scores, the generated answer, and the gold label.

Open item. If the per-question outputs were not saved, I will rerun the pipeline with the same seed before starting.

4.2 Procedure

1. **Baseline.** Fit a single threshold on the top-1 retrieval score to predict “should abstain” (the 258 unanswerable questions), and record accuracy, precision, recall, and F1.
2. **Concept authoring.** Write 6 to 10 concepts as short natural-language prompts. Starting candidates are: the retrieved passage does not mention the question’s main entity; the question asks about something the passage negates; the question contains a false premise; the top passages disagree with each other; and the answer would require information outside the passage.
3. **Sketching.** Combine concepts into 3 to 5 sketch models (single concepts, small sets, all concepts). After each sketch, read 20 misclassified questions and revise or add concepts.
4. **Comparison.** Evaluate every sketch and the baseline on the same 500 questions, then check the best sketch once on a fresh seeded sample of 200 SQuAD 2.0 validation questions.

Open item. ModelSketchBook was built on GPT-3, which is no longer offered. Before starting, I will confirm whether the package runs with a current model and estimate the number of model calls (concepts multiplied by questions).

4.3 Design journal

Throughout steps 2 and 3, I will keep a dated design journal recording each concept added, changed, or dropped, and why. The journal is the data for RQ3.

5. Analysis plan

- **RQ1:** precision and recall of each concept against the abstain label, with example questions each concept catches and misses.
- **RQ2:** abstention F1 for each sketch and the baseline, with bootstrap 95% intervals over the 500 questions; the held-out 200 guard against tuning to the sample.
- **RQ3:** a qualitative coding of journal entries for moments when sketching changed how I framed the problem, such as a new concept, a dropped concept, or a relabeled question.

6. Limitations

- I am the only designer, so RQ3 is a reflective design account, not a user study.
- SQuAD questions are not student questions, and SQuAD passages are not course materials.
- Zero-shot concept scores depend on the underlying model and may not transfer to another one.
- No learners are involved. Any claim about learners’ trust is an implication, not a finding.

7. Declarations

Origin of this work. This study follows up my final project for CSCI E-222, Foundations of Large Language Models, at Harvard University (Spring 2026), reported in Choi (2026).

Use of AI tools. I used Claude (Anthropic), an AI assistant, to verify references against Crossref records and to draft text and tables.

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Competing interests. None.

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